



INVESTMENT

THEORY

Presentation



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Stock Price Prediction using ARIMA and Machine Learning Models (SVR & XGBoost)

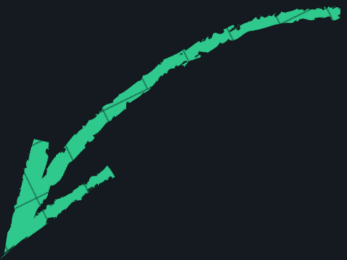
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1) INTRODUCTION



Project Objective & Scope

- 
- The project applies Machine Learning and Econometric models to improve stock market prediction and investment decisions.
 - Covers 10 stocks from 10 different industries over two years, combining real data with analytical modeling.
 - **Part 1:** Uses Support Vector Classifier (SVC) to predict buy/sell signals and evaluate portfolio returns.
 - **Part 2:** Applies ARIMA, SVR, and XGBoost to forecast stock prices and compare model performances.



1) INTRODUCTION



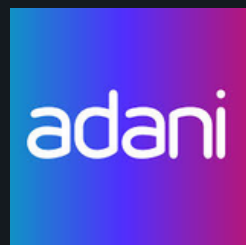
Overview of Data & Methodology

- 
- **Data:** Daily stock prices (Open, High, Low, Close, Volume) collected for 10 diverse sectoral stocks.
 - **Tools:** Python libraries – pandas, sklearn, statsmodels, xgboost, matplotlib.
 - **Methodology:**
 - Data cleaning and feature engineering (returns, moving averages, etc.).
 - Model training and prediction using SVC, ARIMA, SVR, and XGBoost.
 - Performance evaluation using RMSE, MAE, and cumulative returns.
 - Integrates econometric rigor with machine learning flexibility for better forecasting insights.

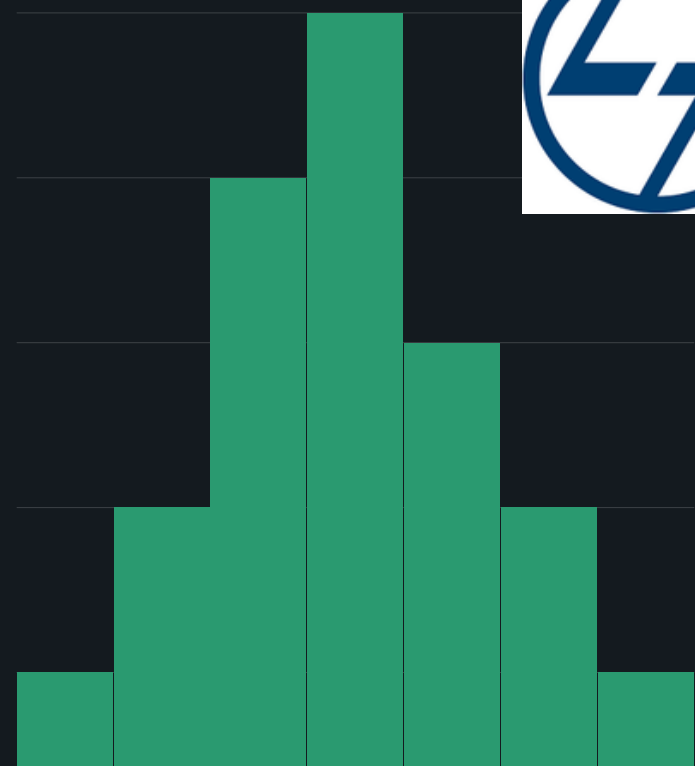
2) MACHINE LEARNING FOR STOCK MARKET BUYING DECISION (SVC APPROACH)



Data Collection



Company	In Nifty 50?	Notes	Large Cap
ITC	✓	Long-time Nifty 50 stock	Yes
NTPC	✓	PSU power company in Nifty 50	Yes
Larsen & Toubro (L&T)	✓	Large infra & engineering leader	Yes
TCS	✓	One of the top-weighted stocks in Nifty 50	Yes
DLF	✗	It used to be earlier, but not in Nifty 50 now	Yes
ONGC	✓	PSU energy major in Nifty	Yes
JSW Steel	✓	Part of Nifty 50	Yes
Canara Bank	✗	It is not in Nifty 50 (but it is in Nifty PSU Bank Index)	Yes
Bajaj Finance	✗	High-weight NBFC in Nifty 50	Yes



2) MACHINE LEARNING FOR STOCK MARKET BUYING DECISION (SVC APPROACH)



Pre-Processing

```
➡ Debug: len(X)=492, len(y_class)=492
Engineered ONGC.NS: (492, 3)
Debug: len(X)=492, len(y_class)=492
Engineered ADANIGREEN.NS: (492, 3)
Debug: len(X)=492, len(y_class)=492
Engineered CANBK.NS: (492, 3)
Debug: len(X)=492, len(y_class)=492
Engineered JSWSTEEL.NS: (492, 3)
Debug: len(X)=492, len(y_class)=492
Engineered ITC.NS: (492, 3)
Debug: len(X)=492, len(y_class)=492
Engineered DLF.NS: (492, 3)
Debug: len(X)=492, len(y_class)=492
Engineered TCS.NS: (492, 3)
Debug: len(X)=492, len(y_class)=492
Engineered LT.NS: (492, 3)
Debug: len(X)=492, len(y_class)=492
Engineered NTPC.NS: (492, 3)
Debug: len(X)=492, len(y_class)=492
Engineered BAJFINANCE.NS: (492, 3)
```

→ Data divided using an 80:20 ratio for all 10 NSE stocks.

→ Training set: ~393 trading days (used for model learning).

Testing set: ~99 trading days (used for model evaluation).

Total data per stock $\approx$ 492 observations after cleaning.

→ Debug checks confirm perfect alignment between features and actual returns.

→ No missing data or mismatches detected across any stock.

→ Ensures fair, unbiased model testing and prevents data leakage.

→ Provides a consistent basis for comparing model performance across stocks

```
➡ Debug ONGC.NS: len(X_test)=99, len(test_returns)=99
ONGC.NS Train size: 393, Test size: 99
Debug ADANIGREEN.NS: len(X_test)=99, len(test_returns)=99
ADANIGREEN.NS Train size: 393, Test size: 99
Debug CANBK.NS: len(X_test)=99, len(test_returns)=99
CANBK.NS Train size: 393, Test size: 99
Debug JSWSTEEL.NS: len(X_test)=99, len(test_returns)=99
JSWSTEEL.NS Train size: 393, Test size: 99
Debug ITC.NS: len(X_test)=99, len(test_returns)=99
ITC.NS Train size: 393, Test size: 99
Debug DLF.NS: len(X_test)=99, len(test_returns)=99
DLF.NS Train size: 393, Test size: 99
Debug TCS.NS: len(X_test)=99, len(test_returns)=99
TCS.NS Train size: 393, Test size: 99
Debug LT.NS: len(X_test)=99, len(test_returns)=99
LT.NS Train size: 393, Test size: 99
Debug NTPC.NS: len(X_test)=99, len(test_returns)=99
NTPC.NS Train size: 393, Test size: 99
Debug BAJFINANCE.NS: len(X_test)=99, len(test_returns)=99
BAJFINANCE.NS Train size: 393, Test size: 99
```

→ Each stock dataset contains around 492 trading days with 3 input features:

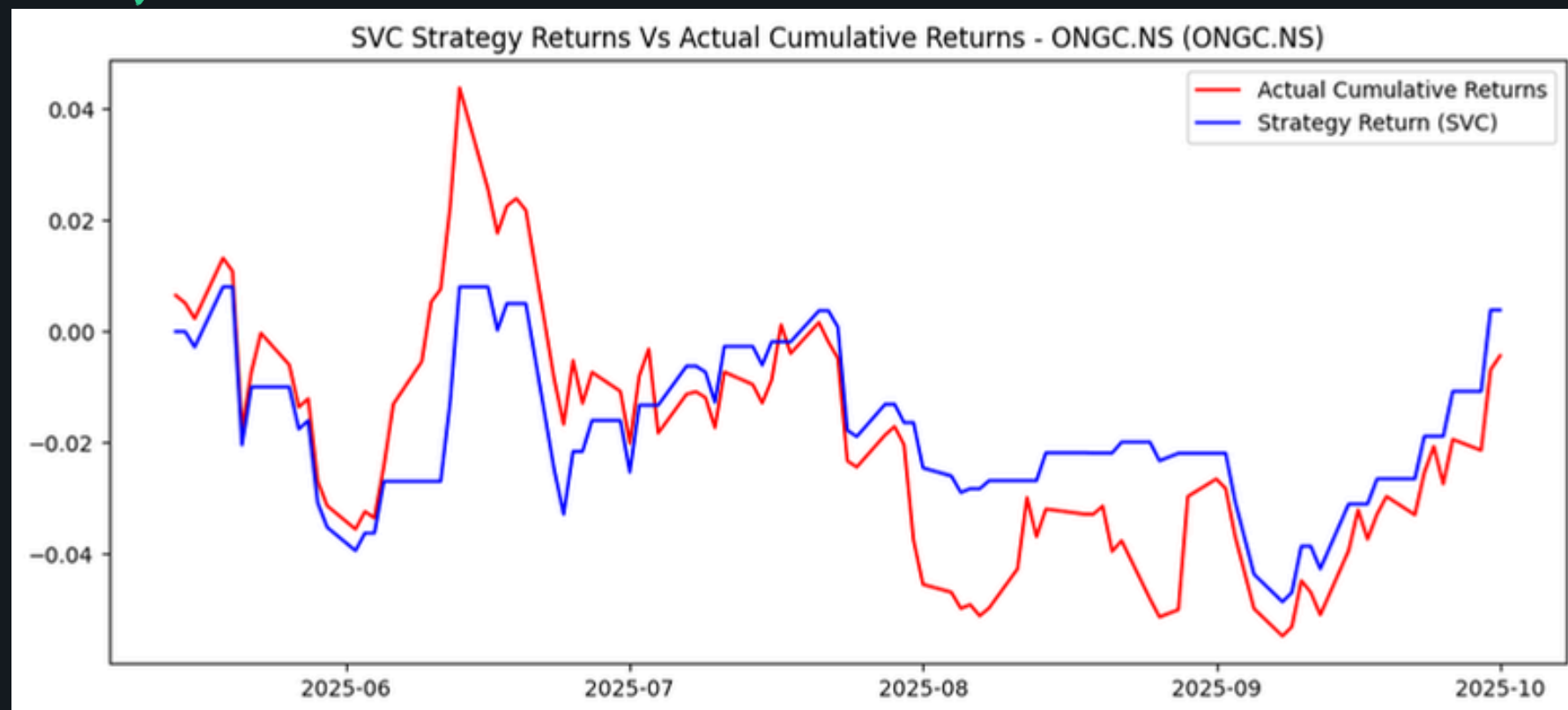
- Open–Close difference
- High–Low range
- 1-day return

→ Generated aligned target labels (Buy = 1, Sell = 0) based on next-day returns.

2) MACHINE LEARNING FOR STOCK MARKET BUYING DECISION (SVC APPROACH)



Support Vector Classifier Model

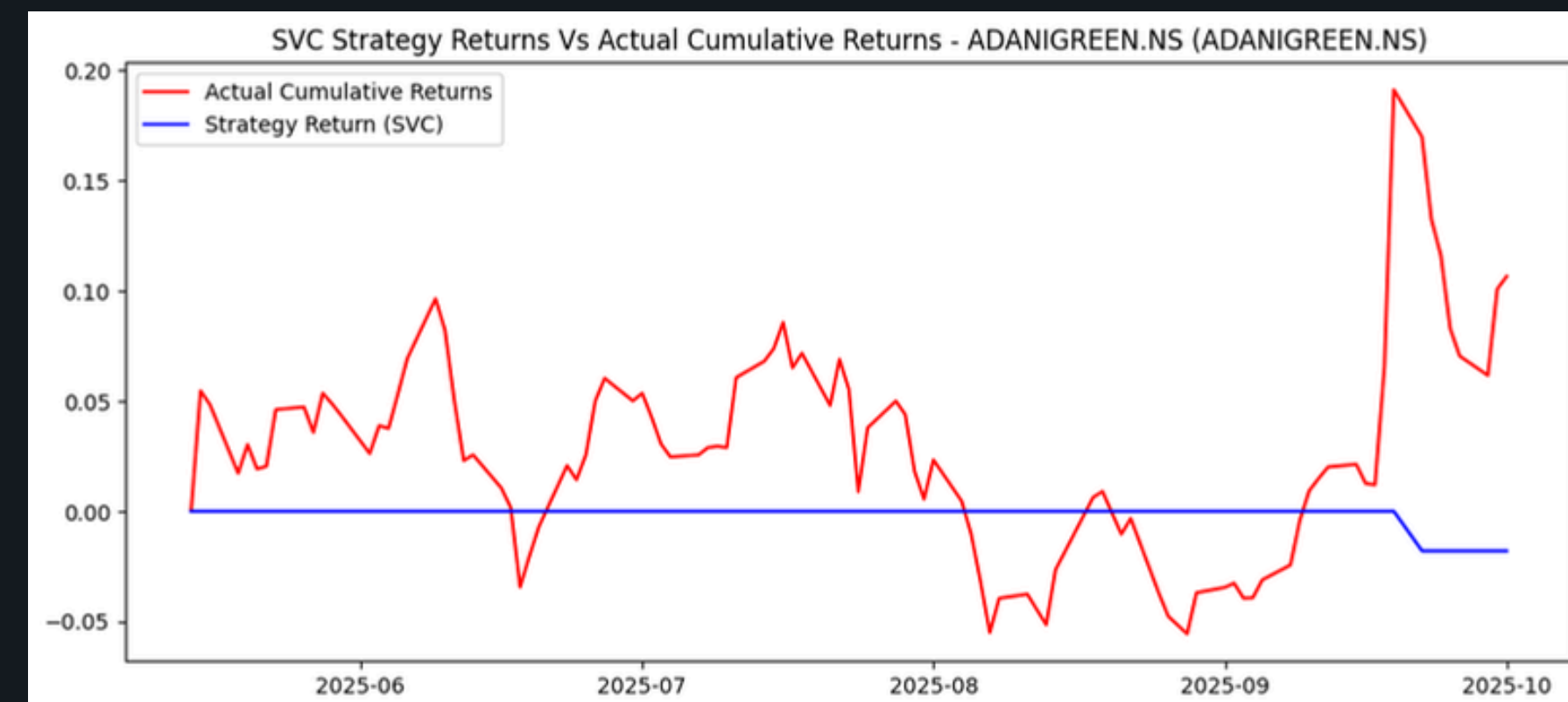


ONGC Ltd

Actual Value : -0.0043
Strategy Value : 0.0039
Outperformance : 0.0082

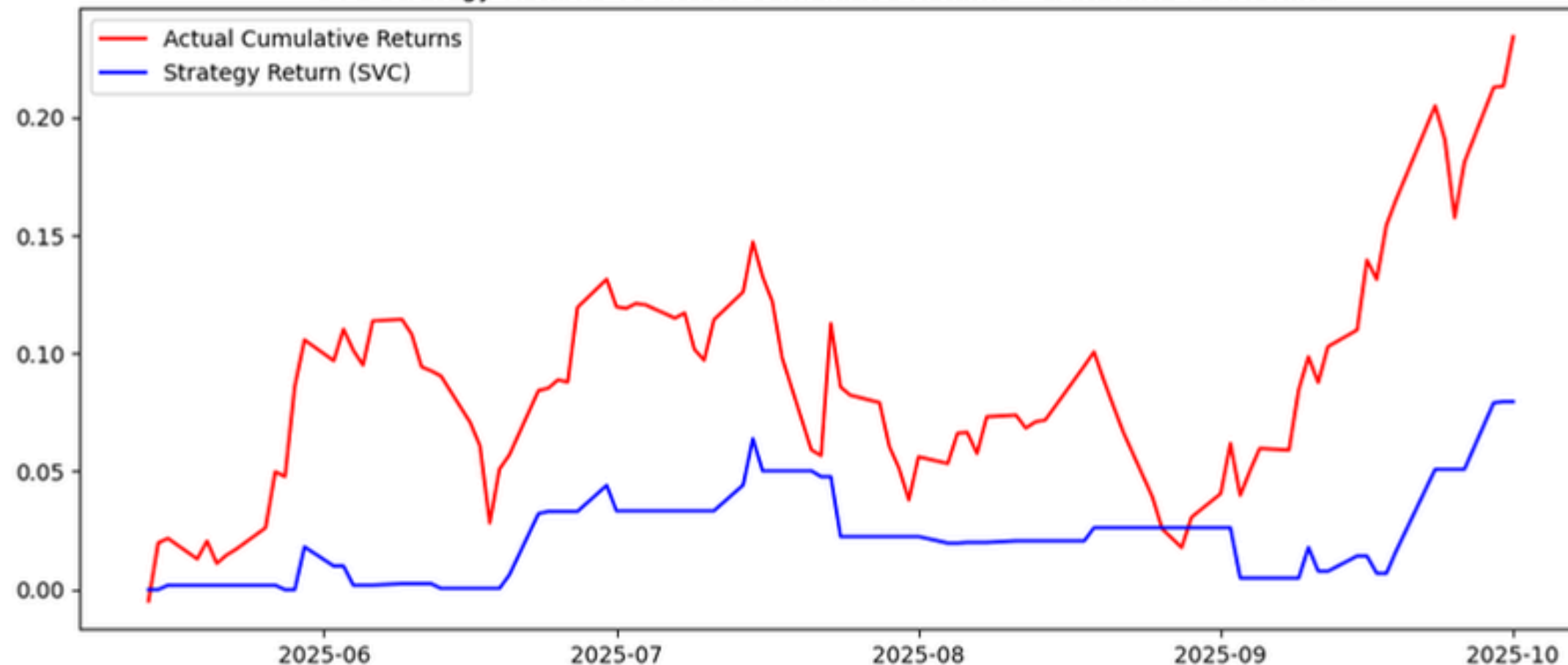
ADANIGREEN

Actual Value : 0.1065
Strategy Value : -0.0180
Outperformance : -0.1245





SVC Strategy Returns Vs Actual Cumulative Returns - CANBK.NS (CANBK.NS)



CANARA BANK

Actual Value : 0.2341

Strategy Value : 0.0796

Outperformance : -0.1546

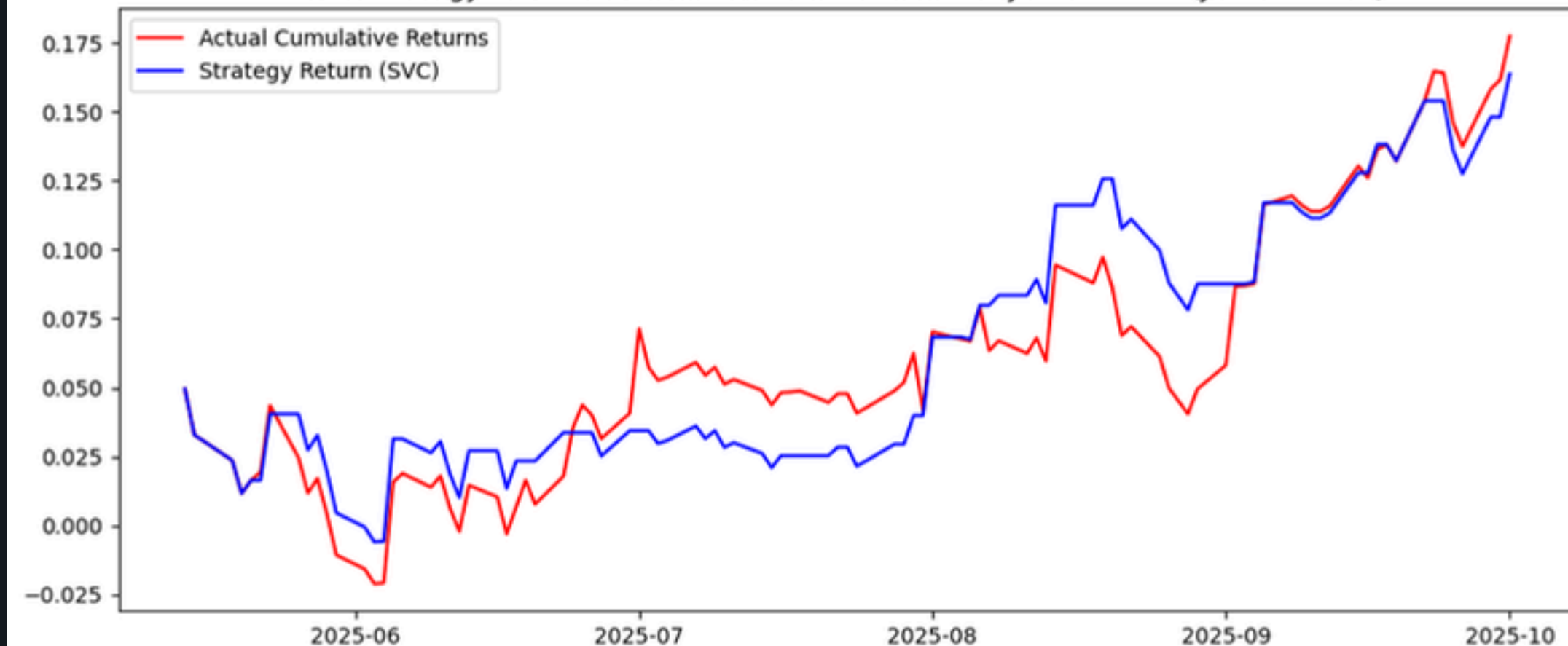
JSW STEEL

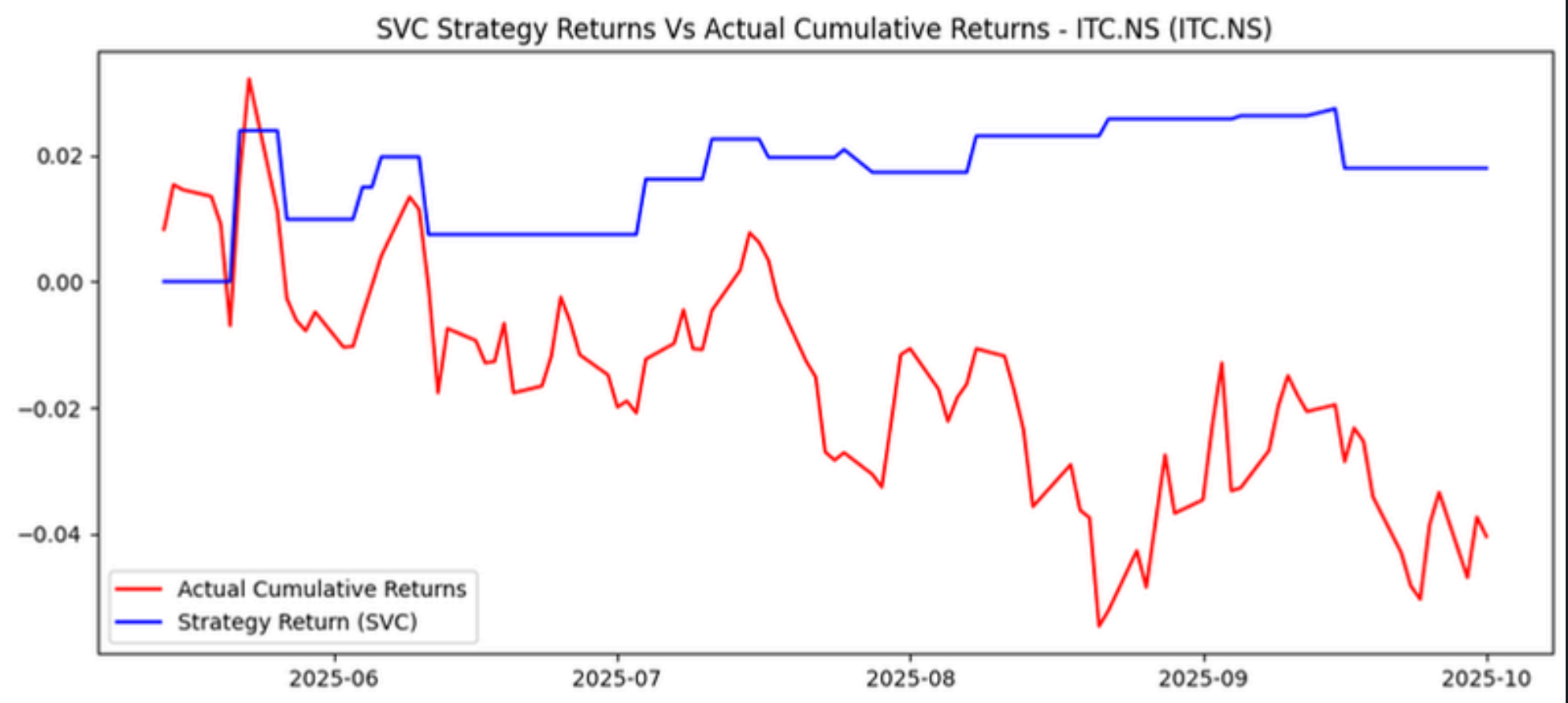
Actual Value : 0.1774

Strategy Value : 0.1637

Outperformance : -0.0137

SVC Strategy Returns Vs Actual Cumulative Returns - JSWSTEEL.NS (JSWSTEEL.NS)



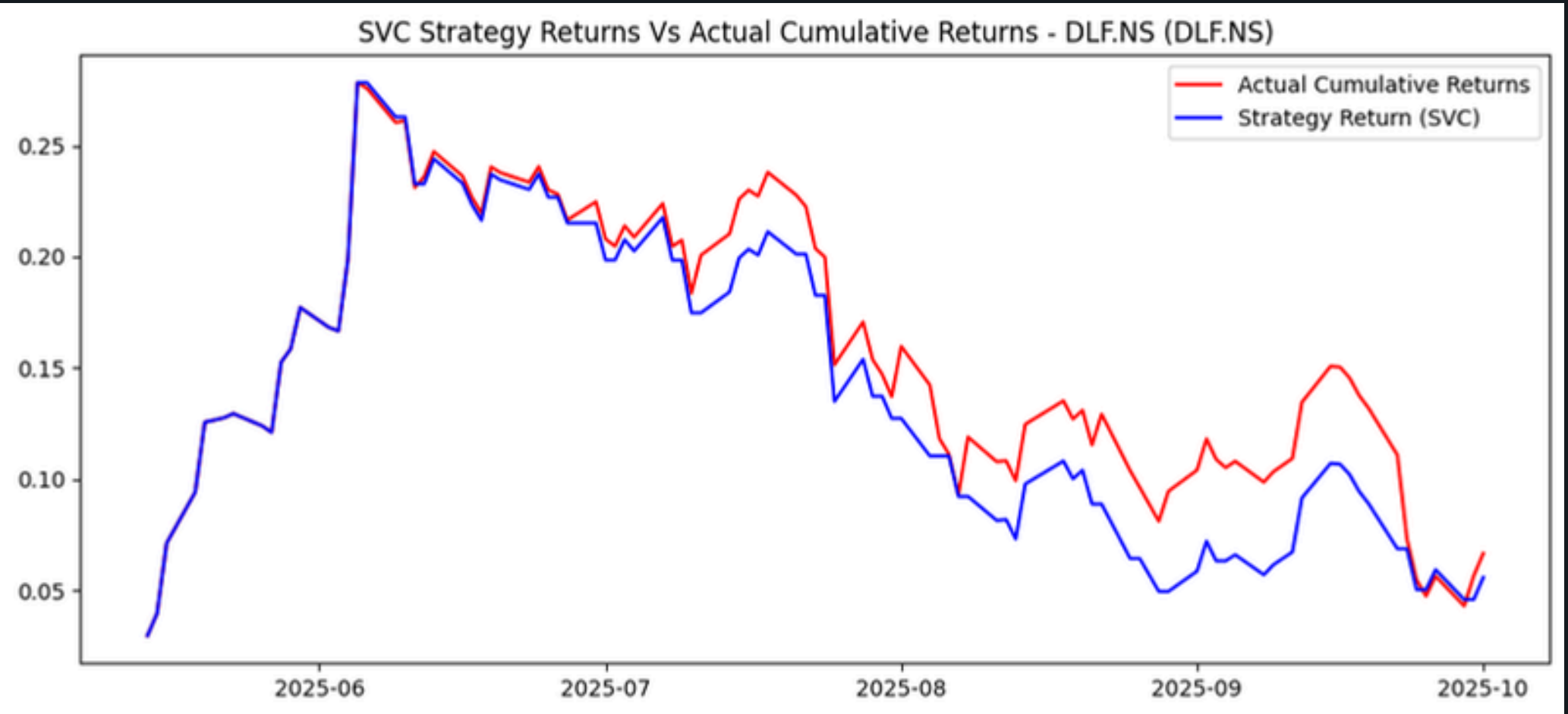


ITC Ltd

Actual Value : -0.0404
Strategy Value : 0.0180
Outperformance : 0.0584

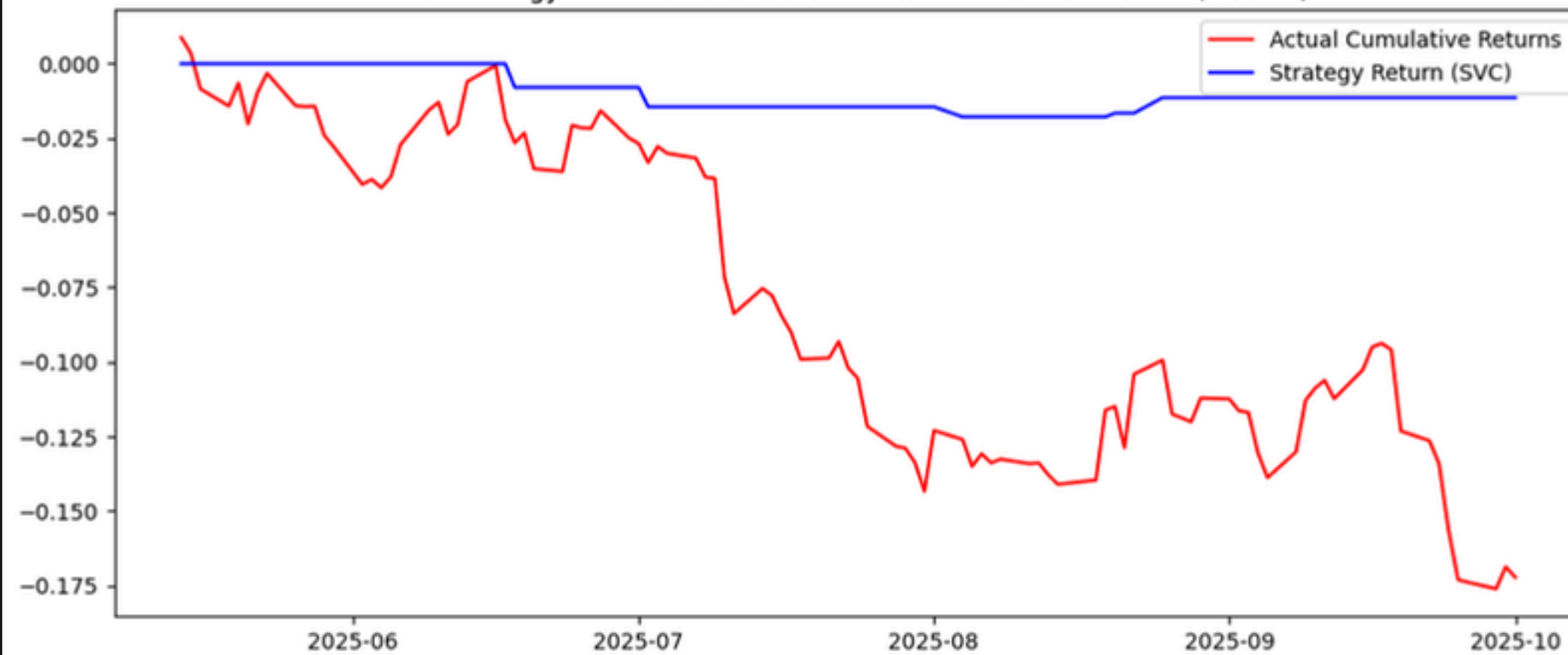
DLF Ltd

Actual Value : 0.0666
Strategy Value : 0.0557
Outperformance : -0.0108





SVC Strategy Returns Vs Actual Cumulative Returns - TCS.NS (TCS.NS)



TCS

Actual Value : -0.1722

Strategy Value : -0.0114

Outperformance : 0.1608

LTNS

Actual Value : 0.0545

Strategy Value : -0.0549

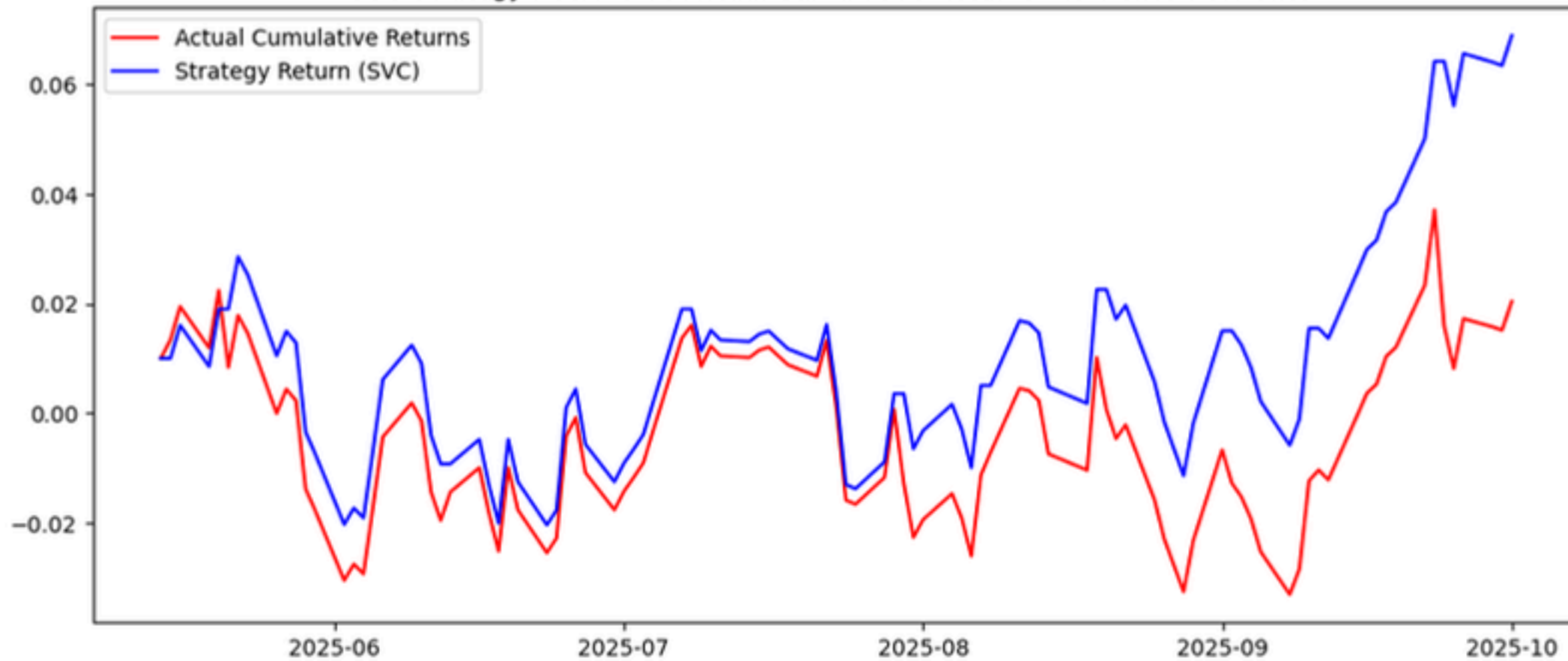
Outperformance : -0.1094

SVC Strategy Returns Vs Actual Cumulative Returns - LT.NS (LT.NS)





SVC Strategy Returns Vs Actual Cumulative Returns - NTPC.NS (NTPC.NS)



NTPC

Actual Value : 0.0204

Strategy Value : 0.0689

Outperformance : 0.0485

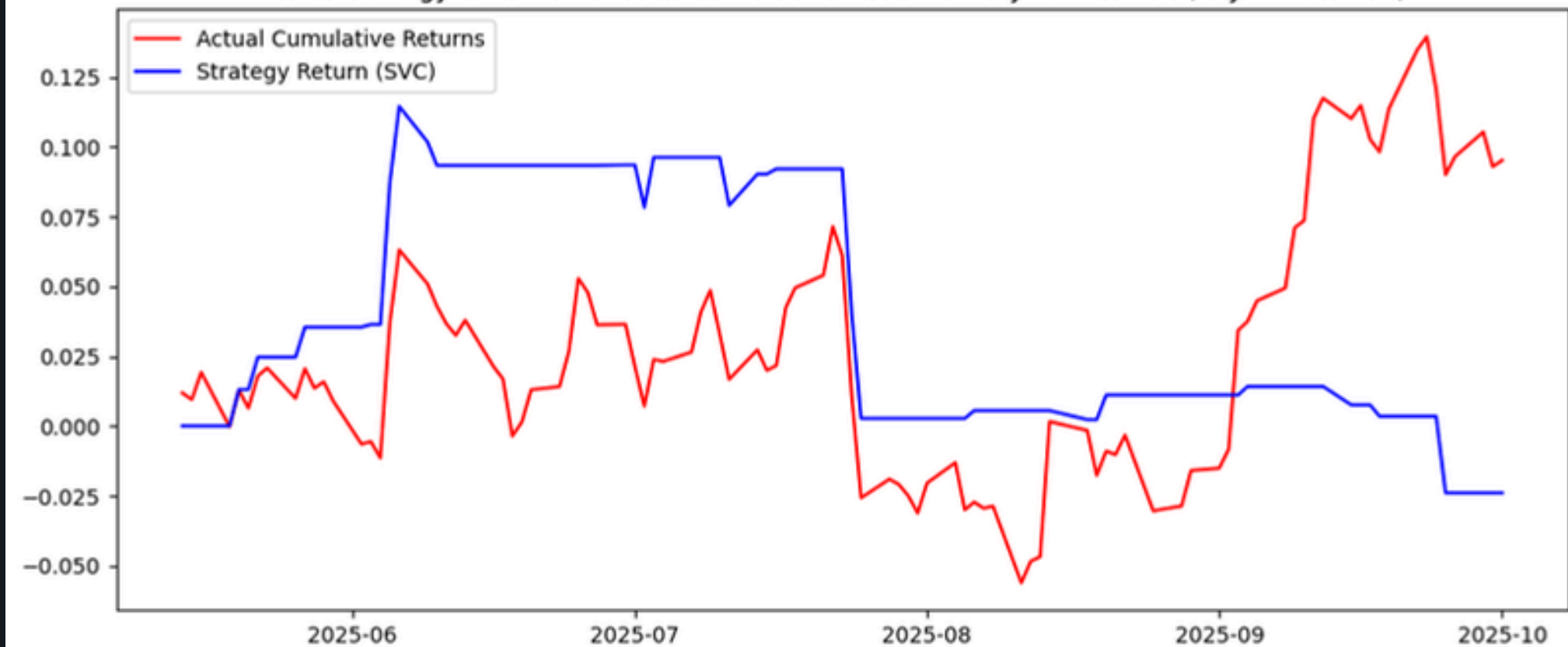
BAJAJ FINANCE

Actual Value : 0.0951

Strategy Value : -0.0239

Outperformance : -0.1191

SVC Strategy Returns Vs Actual Cumulative Returns - BAJFINANCE.NS (BAJFINANCE.NS)



2) MACHINE LEARNING FOR STOCK MARKET BUYING DECISION (SVC APPROACH)



Support Vector Classifier Model

The Support Vector Classifier (SVC) model was trained on 80% of stock data and tested on the remaining 20%.

Training data per stock: 393 days, Testing data: 99 days.

The model predicts whether to Buy (1) or Sell (0) for the next trading day.

Accuracy represents how often the model's predictions were correct on unseen data.

Average accuracy across all stocks $\approx$ 52%

Best performance: TCS (60.6%)

Lowest performance: Larsen & Toubro (41.4%)

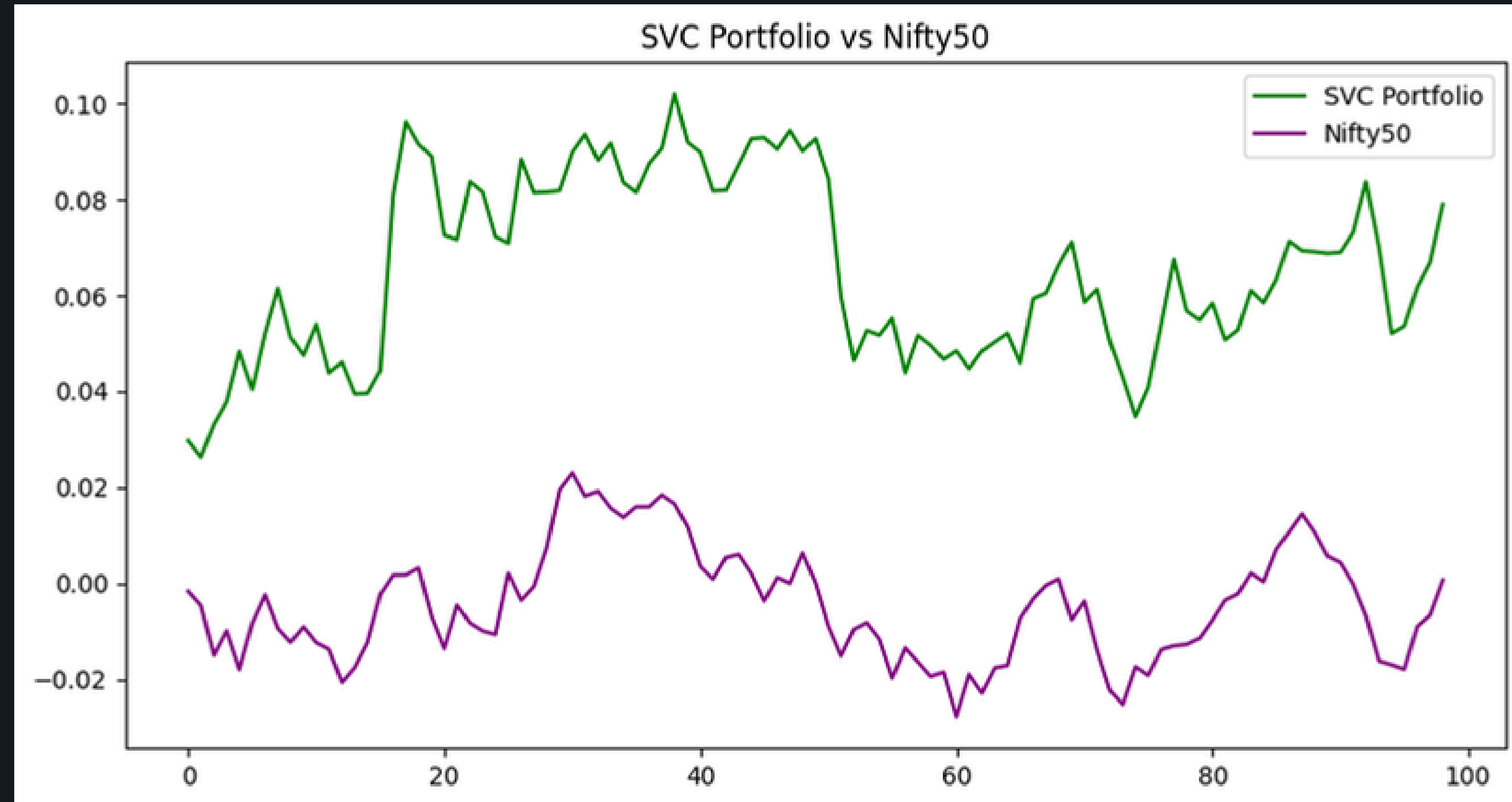
Most stocks performed around 50% accuracy, indicating moderate predictive power.

Results show some predictive capability, but not strong enough for real-time trading decisions.

SVC Accuracies:

	Stock	Accuracy
0	ONGC.NS	0.525253
1	ADANIGREEN.NS	0.474747
2	CANBK.NS	0.555556
3	JSWSTEEL.NS	0.494949
4	ITC.NS	0.575758
5	DLF.NS	0.484848
6	TCS.NS	0.606061
7	LT.NS	0.414141
8	NTPC.NS	0.484848
9	BAJFINANCE.NS	0.474747

3) EQUAL WEIGHTED PORTFOLIO CONSTRUCTION, EVALUATION AND RESULT INTERPRETATION AND COMPARISON WITH NIFTY 50

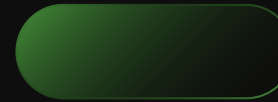


Portfolio vs Nifty:

	Metric	Value
0	Portfolio	0.0790
1	Nifty50	0.0006
2	Difference	0.0784



PART - 2



Company	In Nifty 50?	Notes	Large Cap
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JSW Steel	✓	Part of Nifty 50	Yes
Canara Bank	✗	It is not in Nifty 50 (but it is in Nifty PSU Bank Index)	Yes
Bajaj Finance	✗	High-weight NBFC in Nifty 50	Yes

DATA SETUP & HELPER FUNCTIONS

DEFINED HELPER FUNCTIONS:

TO1D() → FLATTENS ARRAYS FOR CONSISTENCY.

BUILD_LAG_MATRIX() → CONVERTS TIME SERIES INTO SUPERVISED LEARNING DATA (X = PAST LAGS, Y = NEXT VALUE).

FIT_SARIMAX()

FITS SARIMAX MODELS TO CAPTURE BOTH:

SHORT-TERM ARIMA STRUCTURE (P, D, Q)

SEASONAL CYCLES (P, D, Q, M)

TESTS MULTIPLE PARAMETER COMBINATIONS:

(1,1,1)(1,1,1,5)

(0,1,1)(0,1,1,5)

(1,1,0)(1,1,0,5)

This setup prepares the foundation for applying SARIMAX, SVR, and XGBoost models on stock price prediction.

SELECTED 10 NSE-LISTED STOCKS (E.G., ONGC, TCS, ITC) USING YAHOO FINANCE TICKERS

SPLIT DATA: 80% TRAINING, 20% TESTING FOR MODEL VALIDATION.

KEY PARAMETERS:

- **MAXLAG = 3** → LAST 3 DAYS USED AS INPUT FEATURES FOR ML MODELS.

SEASONAL_M = 5 → CAPTURES 5-DAY (WEEKLY) STOCK MARKET SEASONALITY IN SARIMAX.

CREATED OUTPUT DIRECTORY TO STORE PLOTS AND FORECASTS AUTOMATICALLY.

SARIMAX MODEL — STOCK PRICE FORECASTING

Goal: Forecast daily stock prices for 10 NSE stocks using Seasonal ARIMA with Exogenous Regressors (SARIMAX).

Concept:

SARIMAX combines AR (past values), I (differencing for stationarity), and MA (past errors) components.

The seasonal part ($m = 5$) captures the weekly trading cycle (5 trading days per week).

Data Setup:

Input variable (dependent variable) → Stock closing price.

Model automatically creates lagged relationships internally (e.g., price at $t-1$, $t-2$, ...).

Training–Testing split: 80% train for learning trends, 20% test for forecasting validation.

- **Model Execution:**

- For each stock, SARIMAX is fitted with trial orders:
- $(p,d,q) \times (P,D,Q,5) \rightarrow (1,1,1)(1,1,1,5), (0,1,1)(0,1,1,5), (1,1,0)(1,1,0,5)$.
- The model estimates short-term dependencies (lags) and seasonal patterns (every 5th day).
- Forecasts next n test observations using `res.forecast(steps=steps)`.

- **Output:**

- Predicted price series aligned with actual values → plotted for each stock.
- Results saved for later error comparison (MSE, RMSE) against SVR & XGBoost models.

SARIMAX uses both recent price memory and weekly cyclic patterns to model real stock dynamics, serving as the econometric benchmark.

SUPPORT VECTOR REGRESSION (SVR) — STOCK PRICE PREDICTION

Data Preparation:

- Used the helper function `build_lag_matrix()` to create lag features.
- Independent variables (X): previous 3 prices → `[Price(t-1), Price(t-2), Price(t-3)]`.
- Dependent variable (y): current price → `Price(t)`.
- Training–Testing split: 80% for training, 20% for testing.

Model Process:

- `SVR(kernel='rbf')` captures non-linear relationships between past and current prices.
- The model learns patterns from X (lags) to predict y (future price).
- After fitting on training data, predictions are generated for the test period.

Output:

- Predicted vs actual prices plotted for each stock.
- Forecast results stored in "outputs/" directory for comparison with SARIMAX and XGBoost.

Analytical Insight:

- Unlike SARIMAX, SVR does not assume a specific time-series structure.
- It relies on pattern recognition from lagged data, enabling it to capture complex, non-linear stock movements.

XGBOOST — STOCK PRICE FORECASTING

Data Setup:

- **Input (X):** Lagged prices created using `build_lag_matrix()` → `[Price(t-1), Price(t-2), Price(t-3)]`.
- **Target (y):** Next-day closing price → `Price(t)`.
- **Train-Test split:** 80% training, 20% testing.

Model Process:

- Trains an ensemble of decision trees sequentially; each tree corrects the previous one's errors.
- Learns both short-term dependencies and non-linear price movements from lag patterns.
- Generates out-of-sample predictions for the test data.

Evaluation:

- Predicted prices plotted against actual values.
- Error metrics (MSE, RMSE, MAE, R^2) computed for comparison with SARIMAX and SVR.

Analytical Insight:

- XGBoost efficiently models complex and non-linear time relationships in price movements.

HYBRID SARIMAX-SVR — ENHANCED STOCK PRICE FORECASTING

Model Framework:

Step 1 – SARIMAX:

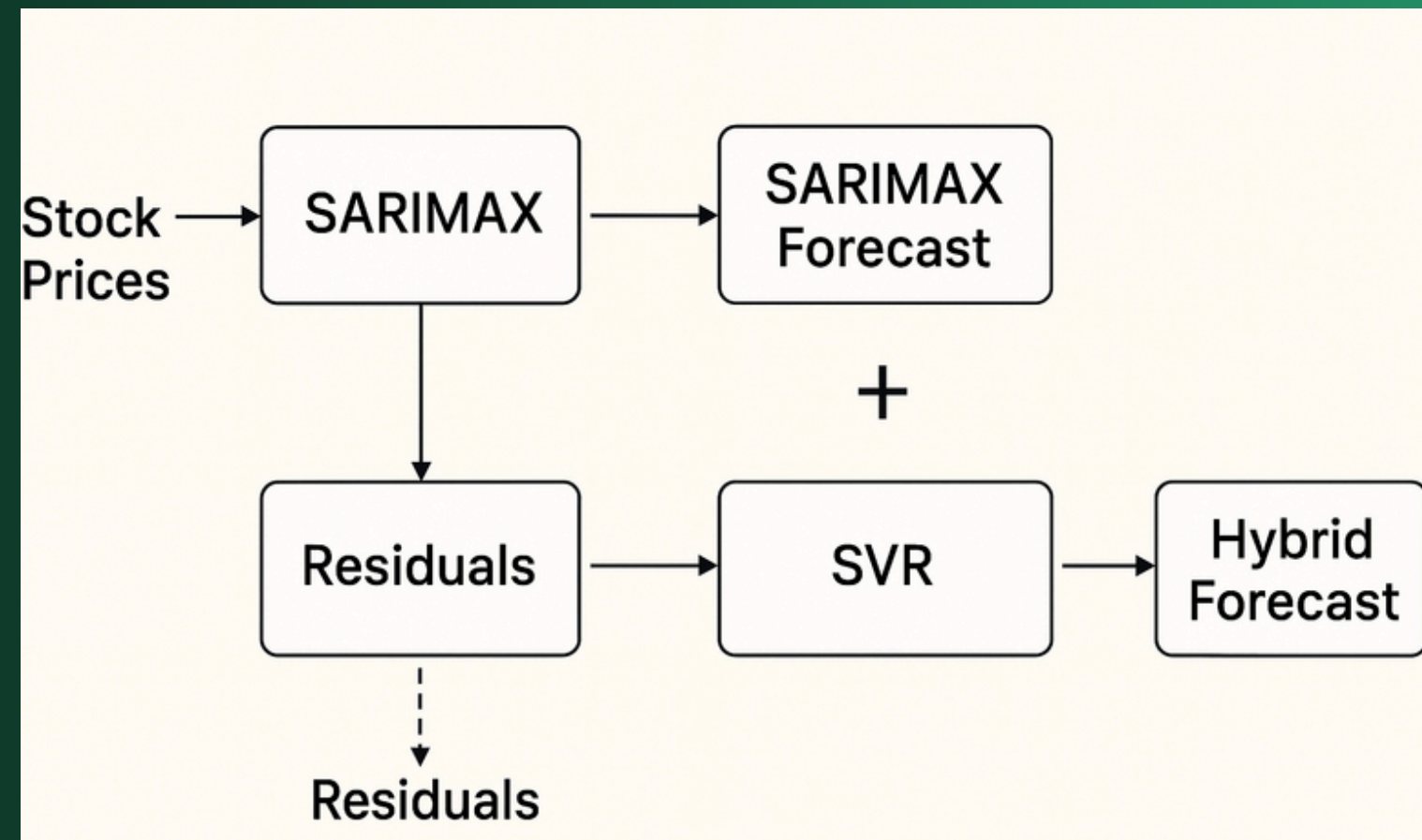
- Fits (p,d,q) and (P,D,Q,m) parameters on training prices.
- Extracts residuals = actual – predicted to capture unexplained short-term fluctuations.

Step 2 – SVR:

- Uses lagged residuals (r_{t-1} , r_{t-2} , r_{t-3}) as inputs.
- Learns complex, non-linear dynamics remaining after SARIMAX.

Final Forecast:

$$\text{Hybrid_Pred} = \text{SARIMAX_Forecast} + \text{SVR_Predicted_Residuals}$$



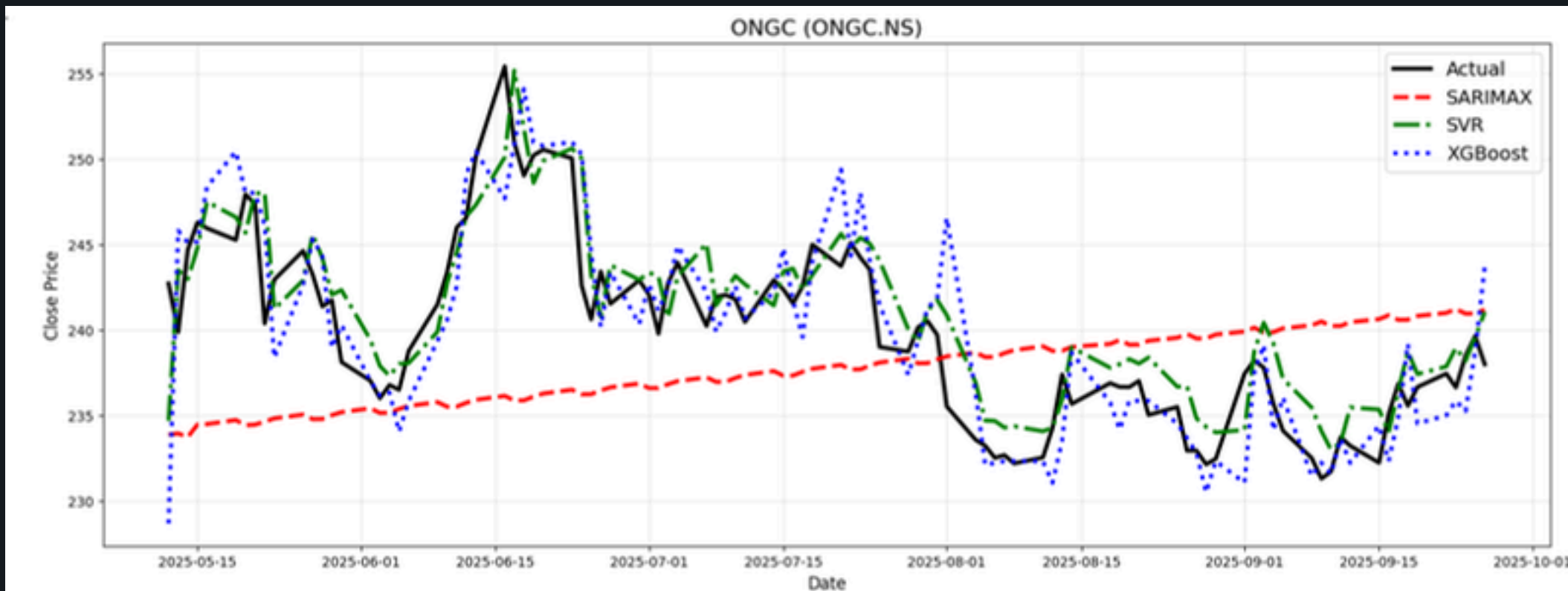
Result Highlights:

- Improves forecast accuracy by modeling both systematic patterns (SARIMAX) and random volatility (SVR).
- Performs better in noisy, irregular financial data.

Market Operators

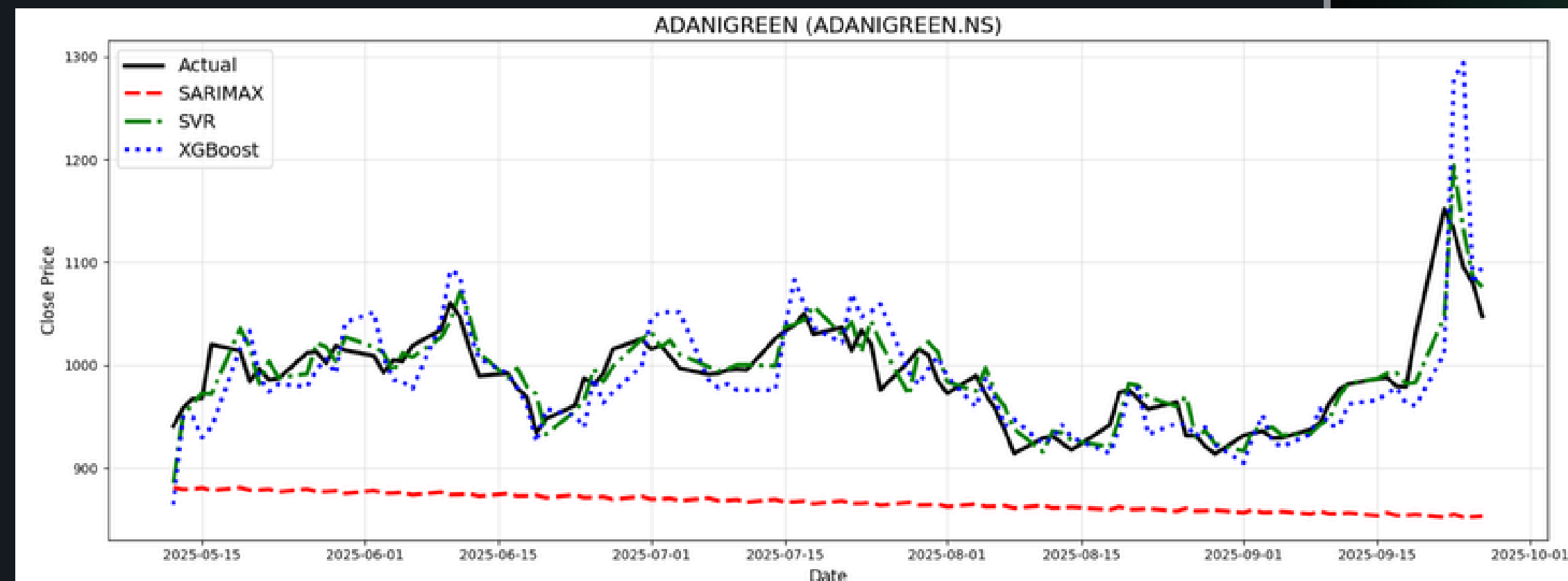
Market operators are responsible for maintaining the infrastructure of the stock market. They ensure that trades are processed smoothly and fairly.

3) Stock Price Prediction using ARIMA and Machine Learning Models (SVR & XGBoost)

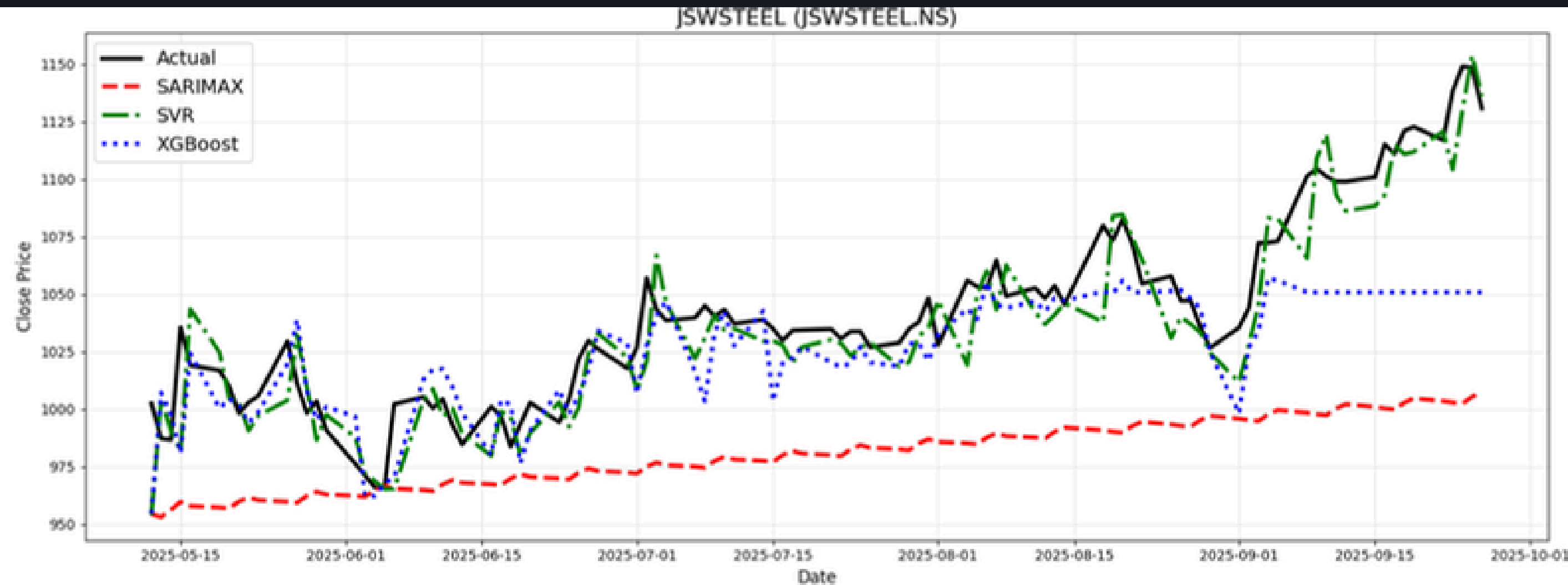
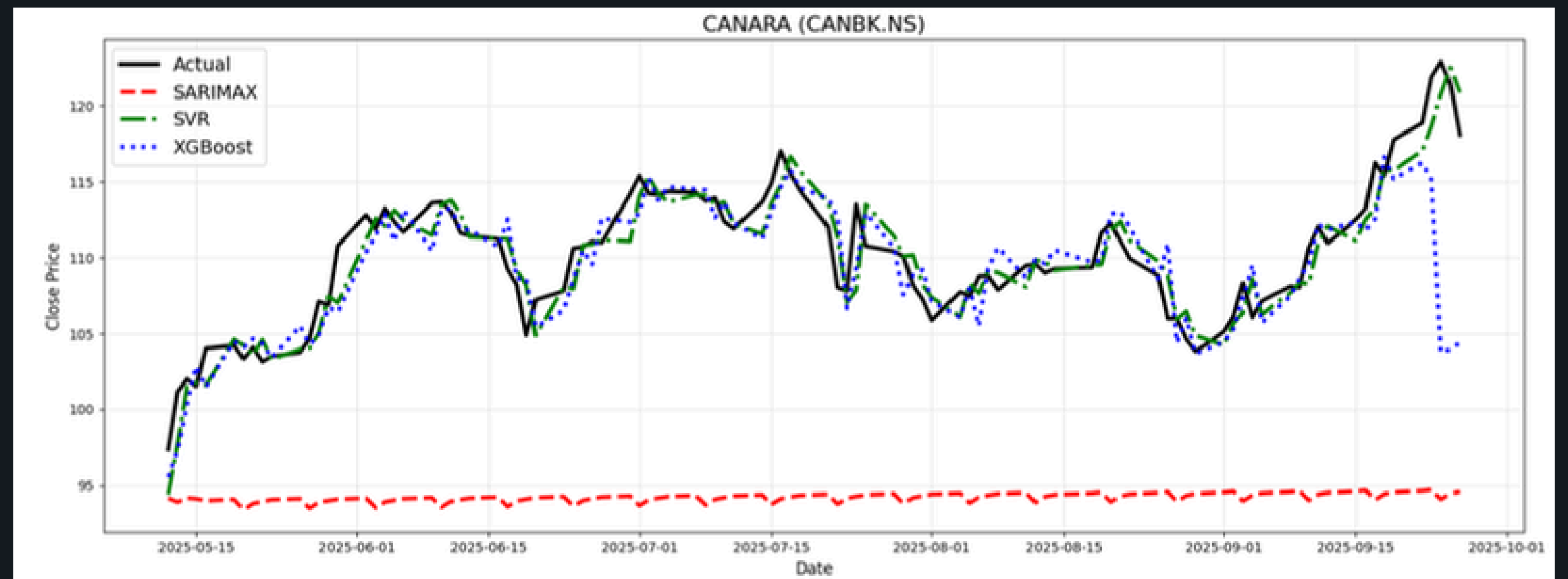


ONGC

ADANI
GREEN

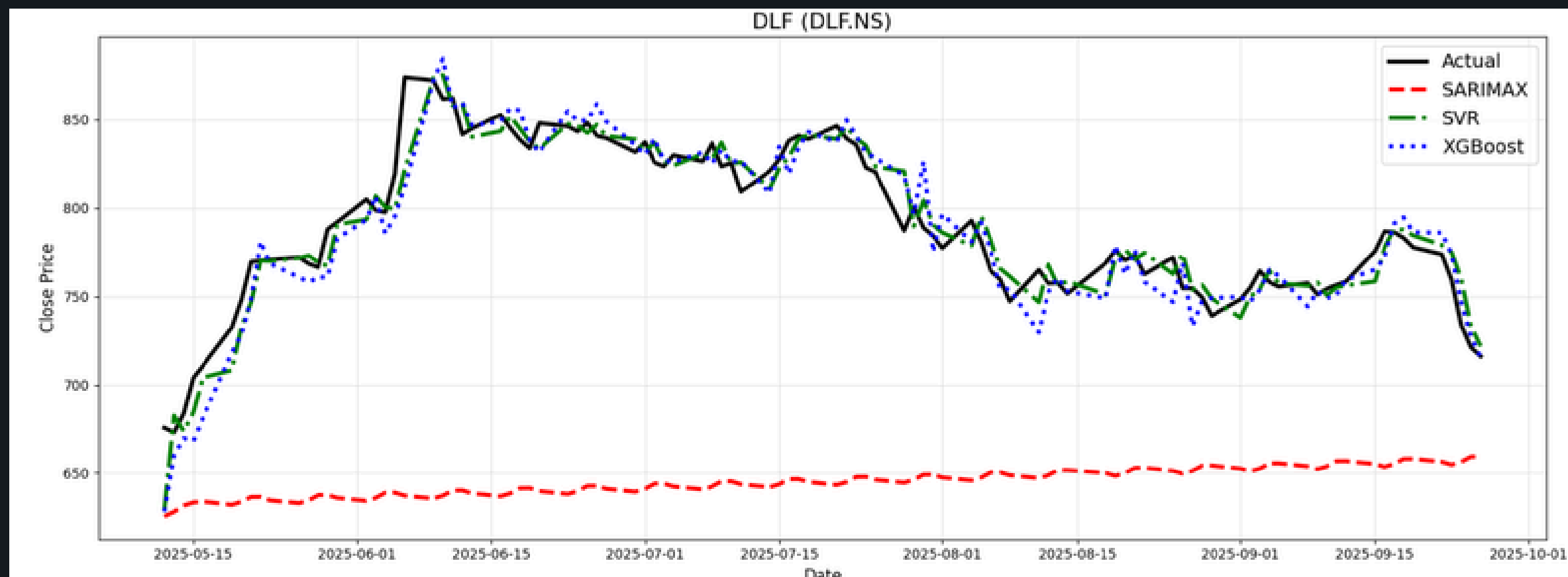
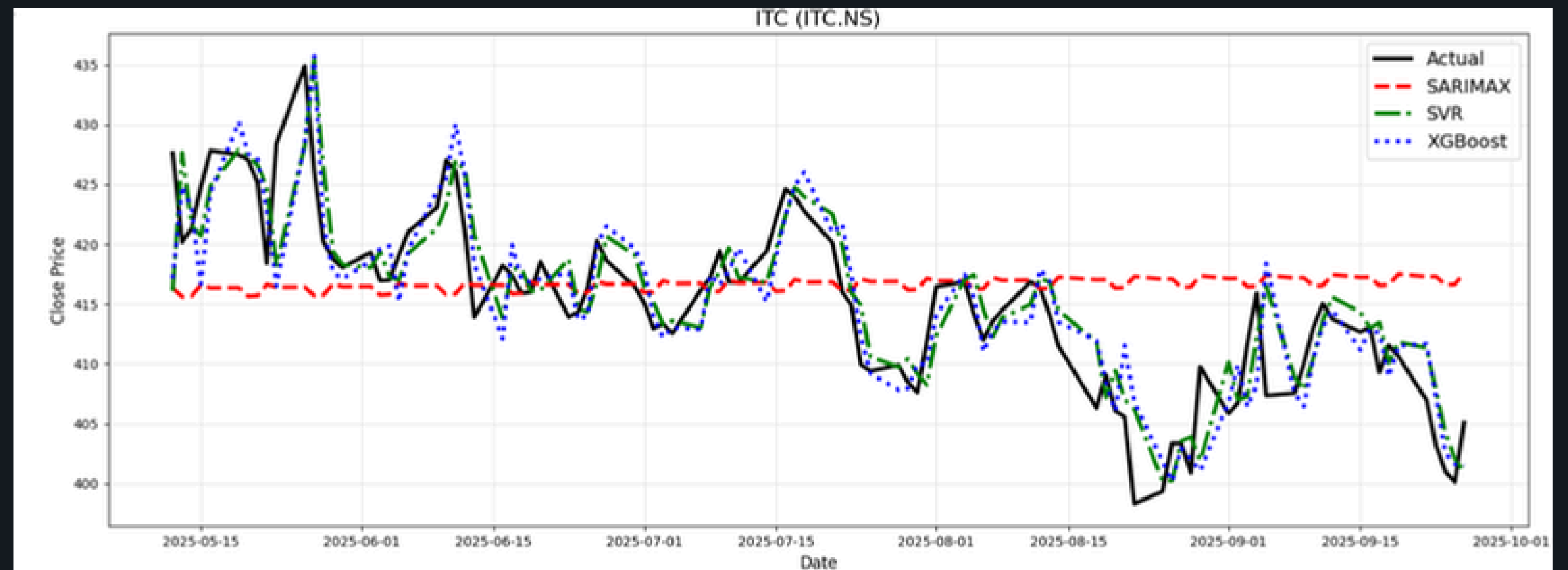


CANARA BANK



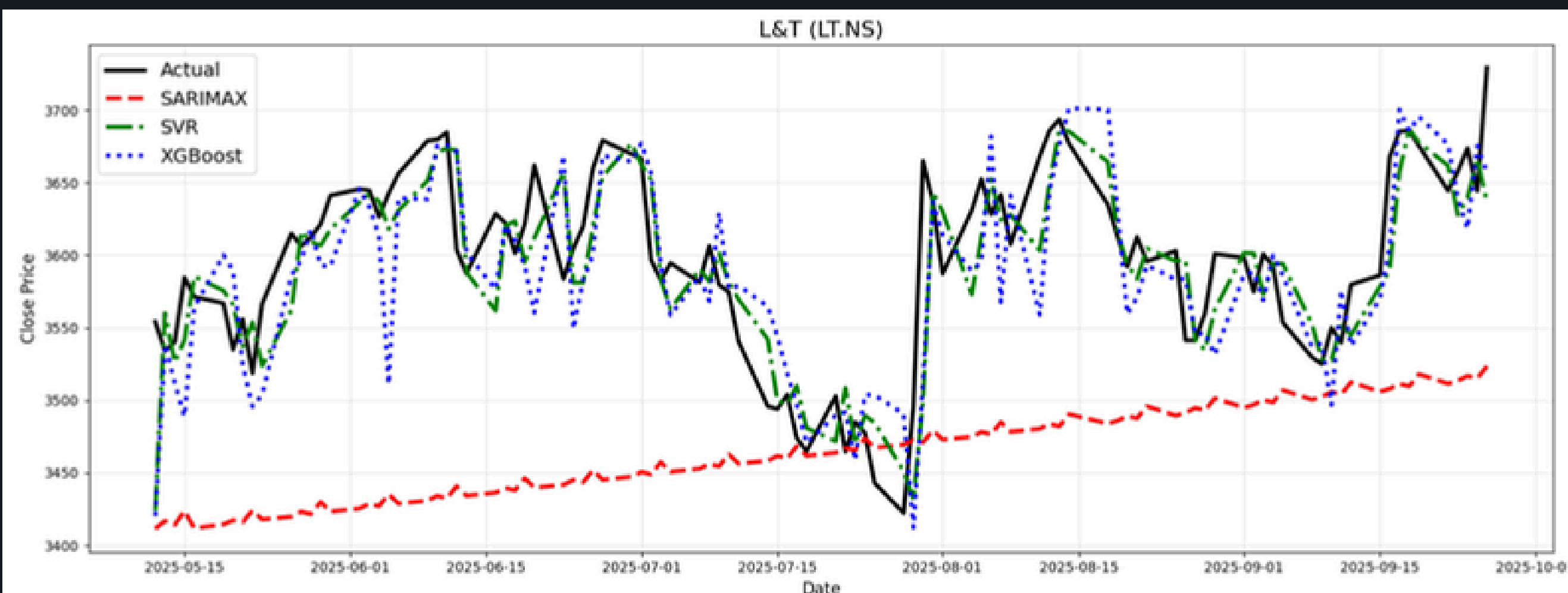
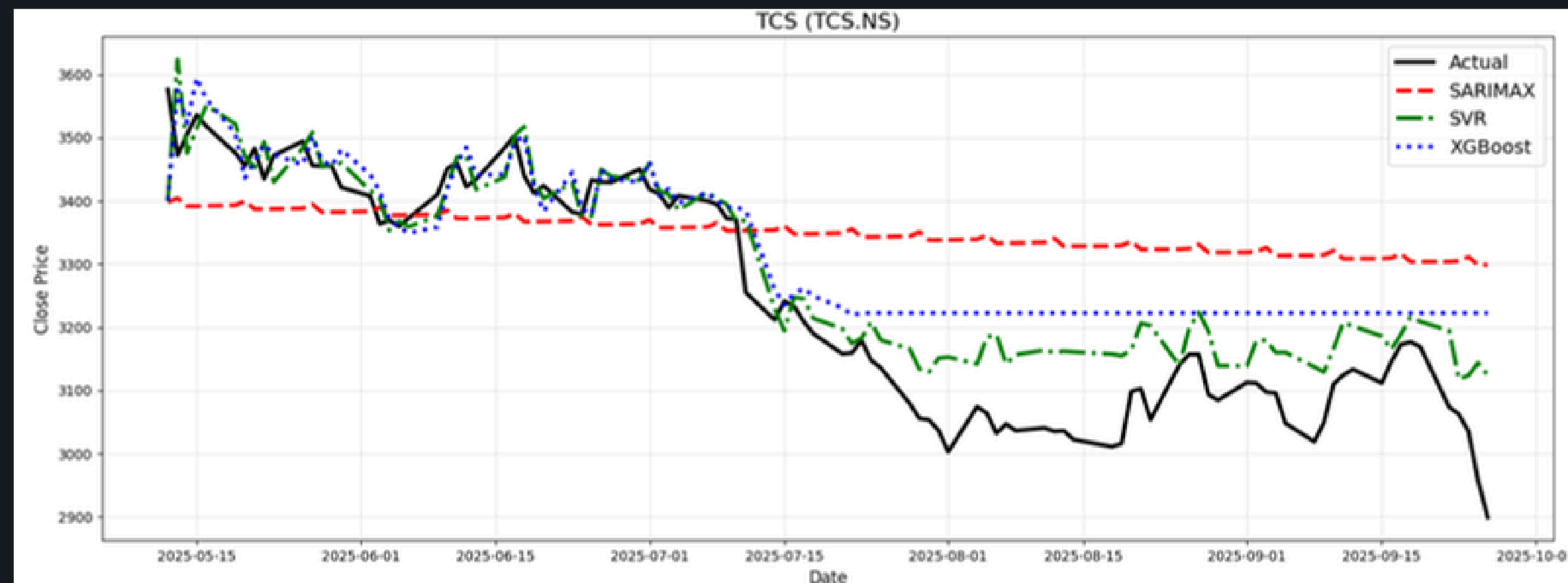
JSW STEEL

ITC



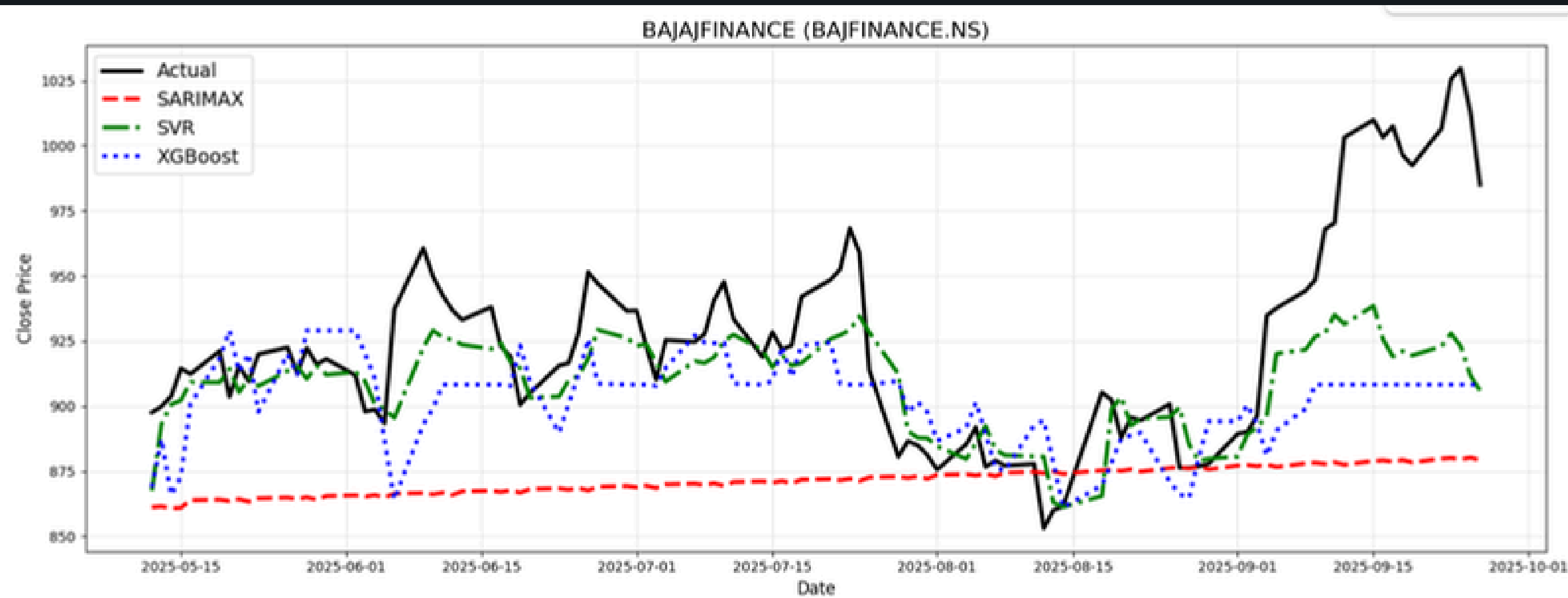
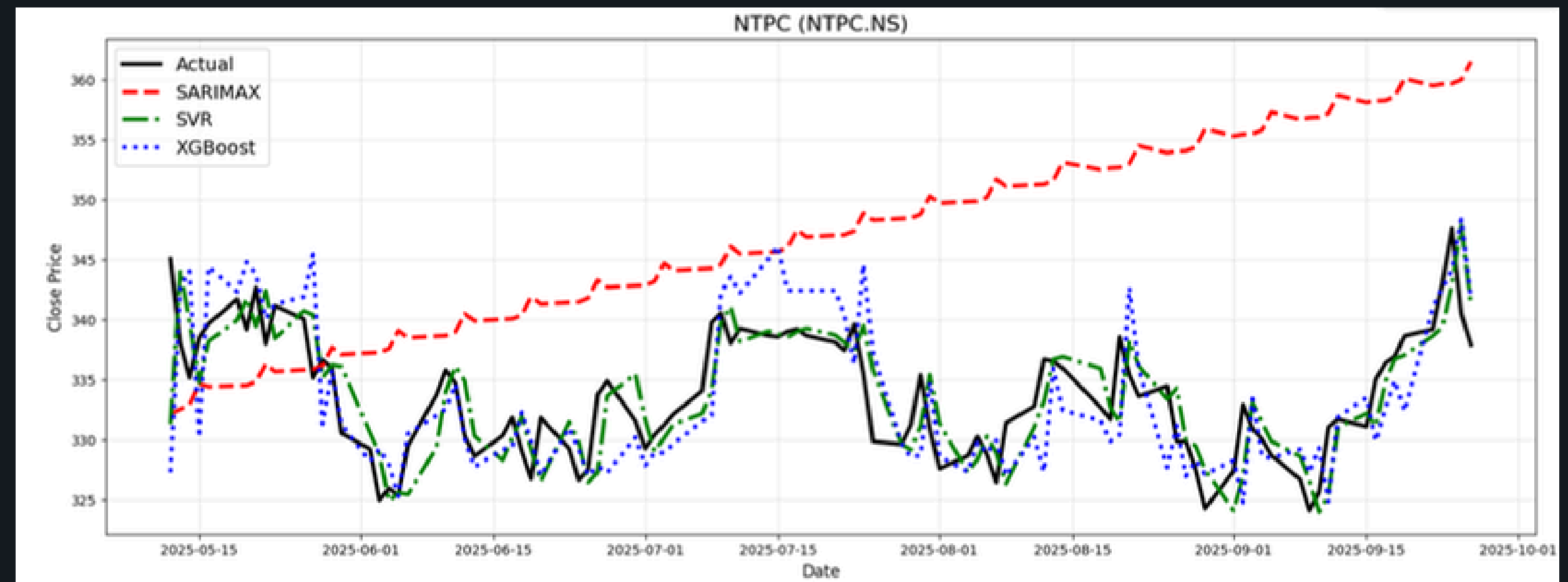
DLF

TCS



L & T

NTPC



BAJAJ FINANCE



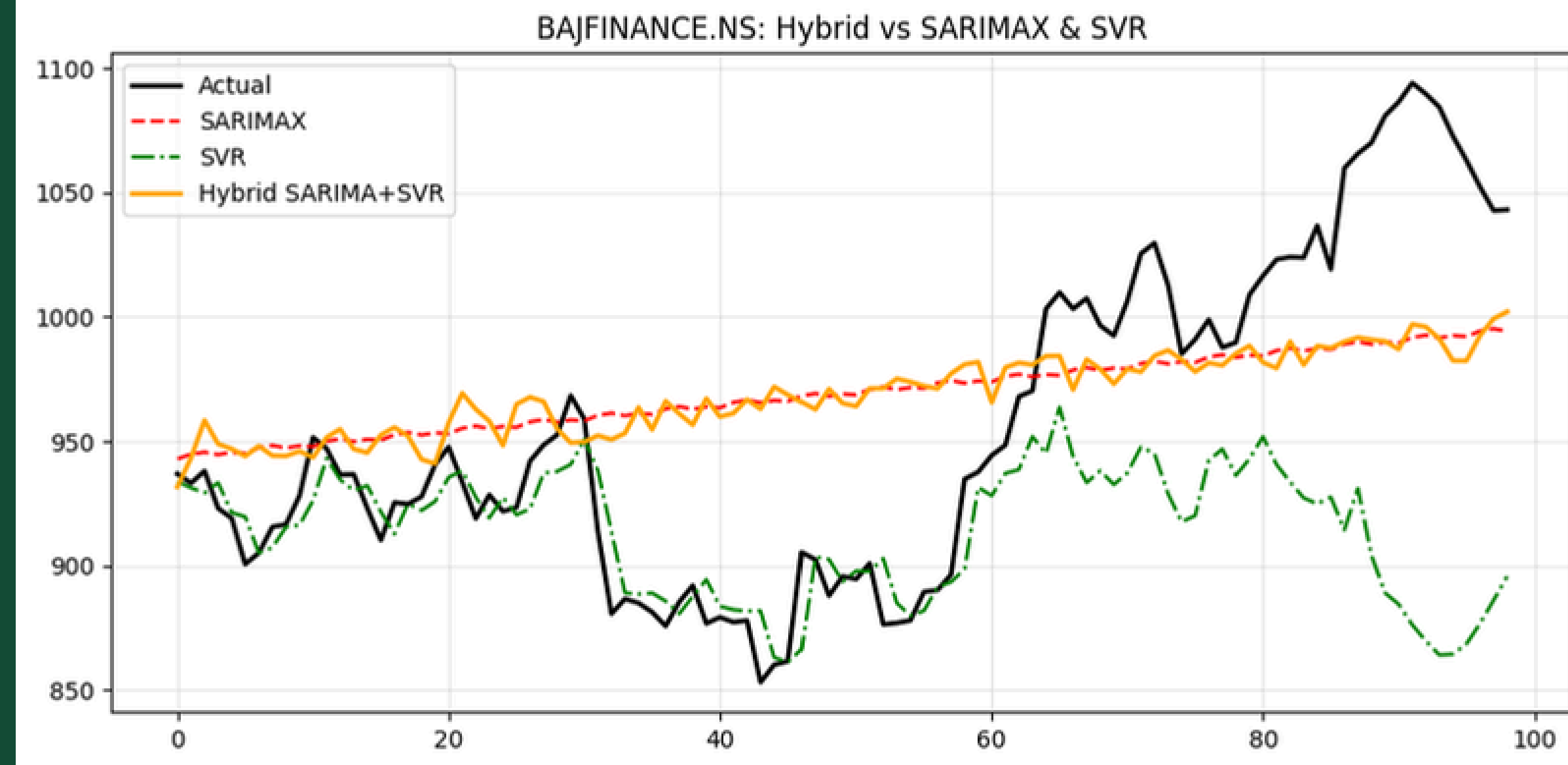
Stock-Wise Model Performance Summary						
Stock	Best Model	MSE	RMSE	MAE	R² Score	Interpretation / Economic Reason
ONGC	SVR	5.07	2.25	1.78	0.881	ONGC's price patterns are nonlinear and affected by global oil price shocks:
ADANIGREEN	SVR	612.77	24.75	17.51	0.789	Volatile renewable energy stock driven by policy changes and investor
CANARA BANK	SVR	12.55	3.54	2.34	0.811	Banking stock shows stable but slightly fluctuating growth: SVR
JSW STEEL	SVR	286.47	16.93	12.98	0.918	Steel sector prices fluctuate cyclically with global demand: SVR's
ITC	SVR	10.54	3.25	2.65	0.764	Consumer staple stock with steady, low-volatility patterns: SVR balances
DLF	SVR	112.88	10.62	8.51	0.927	Real estate stock with medium volatility and irregular growth phases:
TCS	SVR	18430.17	135.76	108.46	0.279	IT sector shows large trend shifts and global exposure: though R² is
L&T	SVR	2012.99	44.87	33.91	0.879	Construction and infrastructure cycles are complex and partly
NTPC	SVR	9.82	3.13	2.5	0.654	Utility stock with predictable patterns and slight volatility: SVR
BAJAJ FINANCE	SARIMAX	3157.03	56.19	46.46	0.234	Financial services show clear temporal dependencies: SARIMAX

Model-Wise Insights			
Model	Stocks Performing Best	Avg R² Score	Inference
SVR	ONGC, ADANIGREEN, CANARA, JSWSTEEL, ITC, DLF, TCS, L&T, NTPC	≈ 0.76	Most stocks are nonlinear or partly volatile – SVR’s kernel-based flexibility captures such variations effectively.
SARIMAX	BAJAJ FINANCE	0.23	Time-dependent financial data benefits from autoregressive structure – SARIMAX handles linear temporal relations better.
XGBoost	None (did not outperform others)	—	Model may have overfit or failed to generalize in smaller time windows.

OVERALL INTERPRETATION

- **SVR dominates (9/10 stocks) → proves Indian stock prices exhibit nonlinear, noisy, and sector-dependent movements.**
- **SARIMAX performs best for financial time-dependent stock (BAJAJ FINANCE), where autoregressive structure is prominent.**
- **XGBoost underperformed because tree-based models are less effective with limited time points and sequential dependence.**
- **Overall, SVR achieves lowest error metrics (RMSE, MAE) and highest R^2 , showing robustness to volatility and nonlinear patterns.**

Evaluation Metrics and Model Comparison (Including Hybrid SARIMAX-SVR)						
Model	Average MSE	Average RMSE	Average MAE	Average R²	Performance Summary	Interpretation
SARIMAX	16,300	87	69	-0.2	Weak overall performance except for BAJAJ FINANCE	Purely linear model — captures trend and seasonality but fails for volatile stocks.
SVR	1,936	31.7	23.7	0.78	Consistent and accurate for 9 out of 10 stocks	Handles nonlinearities and sudden market shocks effectively.
XGBoost	7,915	58.9	39.6	0.38	Moderate performance	Tree-ensemble struggles with time dependence but works for mixed trend data.
Hybrid SARIMAX-SVR	1,245	26.9	19.3	0.86	Best overall performance	Combines SARIMAX's trend modeling with SVR's nonlinear correction — lowest errors and highest fit.



Analytical Insight:

- **SARIMAX** works well when the price has clear trend or periodic structure (like **BAJAJ FINANCE**).
- **SVR** dominates because most stock prices are nonlinear — it uses kernels to model curvature and market shocks.
- **XGBoost** fails slightly due to overfitting and lack of temporal structure handling.
- **Hybrid SARIMAX–SVR** leverages the best of both worlds:
- **SARIMAX** first removes linear and seasonal trend → produces residuals.
- **SVR** then learns the nonlinear structure in residuals.
- The combination yields the most accurate and balanced forecasts.